Week 9: Sales and Customer Analytics System

The e-commerce company wants to analyze its sales data to understand customer purchasing behavior, product performance, and revenue trends. The management team needs reports to identify top customers, best-selling products, and profitable categories.

Generate Total Sales Reports

1. Total number of orders placed.

```
SQL> SELECT COUNT(*) AS Number_Of_Orders_Placed FROM Orders;

NUMBER_OF_ORDERS_PLACED
-----------------------
                     10
```

2. Total revenue generated from sales.

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS Total_Revenue FROM Orders;

TOTAL_REVENUE
-------------
        59900
```

3. Average order value.

```
SQL> SELECT ROUND(AVG(TOTAL_AMOUNT),2) AS Average_Order_Value FROM Orders;

AVERAGE_ORDER_VALUE
-------------------
               5990
```

4. Highest and lowest order amount.

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS Highest_Order,
  2         MIN(TOTAL_AMOUNT) AS Lowest_Order
  3  FROM Orders;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
        18000         2400
```

5. Total sales for a specific period.

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS Total_Revenue
  2  FROM Orders
  3  WHERE ORDER_DATE BETWEEN DATE '2026-09-20'
  4                       AND DATE '2026-09-26';

TOTAL_REVENUE
-------------
        59900
```

Customer Purchase Analysis

1. Total number of orders placed by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID          CUSTOMER_NAME                                TOTAL_ORDERS
-------------------- ---------------------------------------- ------------
C-10001              Ricky Ponting                                          10
C-10006              Vikram Reddy                                            1
C-10003              Nithin Murugan                                          1
C-10008              Suresh Nair                                             1
C-10011              Divya Pillai                                            1
C-10009              Kavya Rajan                                             1
C-10005              Ananya Iyer                                             1
C-10010              Karthik Subramanian                                     1
C-10004              Rahul Menon                                             1
C-10002              Arjun Kumar                                             1
C-10007              Meera Krishnan                                          1

11 rows selected.
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      NVL(SUM(O.TOTAL_AMOUNT), 0) AS AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY AMOUNT_SPENT DESC;

CUSTOMER_ID          CUSTOMER_NAME                          AMOUNT_SPENT
-------------------- -------------------------------------- ------------
C-10001              Ricky Ponting                                 59900
C-10007              Meera Krishnan                                 3000
C-10009              Kavya Rajan                                    2800
C-10005              Ananya Iyer                                    2500
C-10003              Nithin Murugan                                 2200
C-10011              Divya Pillai                                   2100
C-10010              Karthik Subramanian                            1900
C-10004              Rahul Menon                                    1800
C-10008              Suresh Nair                                    1700
C-10002              Arjun Kumar                                    1500
C-10006              Vikram Reddy                                   1200

11 rows selected.
```

3. Average spending per customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      ROUND(NVL(AVG(O.TOTAL_AMOUNT), 0), 2) AS AVERAGE_SPENDING
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_ID          CUSTOMER_NAME                          AVERAGE_SPENDING
-------------------- -------------------------------------- ----------------
C-10001              Ricky Ponting                                      5990
C-10007              Meera Krishnan                                     3000
C-10009              Kavya Rajan                                        2800
C-10005              Ananya Iyer                                        2500
C-10003              Nithin Murugan                                     2200
C-10011              Divya Pillai                                       2100
C-10010              Karthik Subramanian                                1900
C-10004              Rahul Menon                                        1800
C-10008              Suresh Nair                                        1700
C-10002              Arjun Kumar                                        1500
C-10006              Vikram Reddy                                       1200

11 rows selected.
```

4. Customers with maximum purchase amount.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTAL_AMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMER_ID
 18      )
 19  );

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
C-10001              Ricky Ponting
               59900
```

5. Customers with fewer purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDER_ID) <= 1
 13  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_ID          CUSTOMER_NAME                            NUMBER_OF_ORDERS
-------------------- ---------------------------------------- ----------------
C-10002              Arjun Kumar                                             1
C-10006              Vikram Reddy                                            1
C-10003              Nithin Murugan                                          1
C-10008              Suresh Nair                                             1
C-10009              Kavya Rajan                                             1
C-10007              Meera Krishnan                                          1
C-10005              Ananya Iyer                                             1
C-10010              Karthik Subramanian                                     1
C-10004              Rahul Menon                                             1
C-10011              Divya Pillai                                            1

10 rows selected.
```

Find Top Customers Based on Purchase Amount

2. Customers with maximum number of orders.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  5      SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  HAVING SUM(O.TOTAL_AMOUNT) > 20000
 14  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID          CUSTOMER_NAME                          NUMBER_OF_ORDERS
-------------------- -------------------------------------- ----------------
TOTAL_SPENDING
--------------
C-10001              Ricky Ponting                                    10
         59900
```

Identify Best-Selling Products

1. Products with maximum sales quantity.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDER_DETAILS OD
  7      ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID   PRODUCT_NAME                    TOTAL_QUANTITY_SOLD
------------ ------------------------------ -------------------
P100004      Gaming Controller                                 6
P100008      Power Bank                                        2
P100001      Gaming Headset                                    2
P100003      Mechanical Keyboard                               2
P100002      Wireless Mouse                                    2
P100006      Bluetooth Speaker                                 1
P100007      Gaming Monitor                                    1

7 rows selected.
```

2. Products generating highest revenue.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  5      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM PRODUCT P
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_ID   PRODUCT_NAME                    QUANTITY_SOLD TOTAL_REVENUE
------------ ------------------------------ ------------- -------------
P100007      Gaming Monitor                             1         18000
P100004      Gaming Controller                          6         16800
P100003      Mechanical Keyboard                        2          7000
P100001      Gaming Headset                             2          5000
P100006      Bluetooth Speaker                          1          4500
P100008      Power Bank                                 2          3600
P100002      Wireless Mouse                             2          2400

7 rows selected.
```

3. Least-selling products.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  5  FROM PRODUCT P
  6  LEFT JOIN ORDER_DETAILS OD
  7      ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY QUANTITY_SOLD ASC;

PRODUCT_ID   PRODUCT_NAME                  QUANTITY_SOLD
------------ ------------------------------ -------------
P100005      Smartphone                                 0
P100007      Gaming Monitor                             1
P100006      Bluetooth Speaker                          1
P100002      Wireless Mouse                             2
P100003      Mechanical Keyboard                        2
P100008      Power Bank                                 2
P100001      Gaming Headset                             2
P100004      Gaming Controller                          6

8 rows selected.
```

4. Products requiring promotion.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      P.PRICE,
  5      P.STOCK_QUANTITY,
  6      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  7  FROM PRODUCT P
  8  LEFT JOIN ORDER_DETAILS OD
  9      ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY
 11      P.PRODUCT_ID,
 12      P.PRODUCT_NAME,
 13      P.PRICE,
 14      P.STOCK_QUANTITY
 15  HAVING NVL(SUM(OD.QUANTITY), 0) <= 1
 16      AND P.STOCK_QUANTITY > 0
 17  ORDER BY QUANTITY_SOLD, P.PRODUCT_NAME;

PRODUCT_ID   PRODUCT_NAME                    PRICE STOCK_QUANTITY QUANTITY_SOLD
------------ ------------------------------ ----------- -------------- -------------
P100005      Smartphone                     25000               8             0
P100006      Bluetooth Speaker               4500              18             1
P100007      Gaming Monitor                 18000               6             1
```

Perform Category-wise Sales Analysis

1. Total sales generated by each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME                                                                                                    TOTAL_SALES
---------------------------------------------------------------------------------------------------------------- -----------
Gaming                                                                                                             34800
Electronics                                                                                                         9500
Accessories                                                                                                         9400
Mobiles                                                                                                             3600
```

2. Number of products sold in each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY PRODUCTS_SOLD DESC;

CATEGORY_NAME                                                                                                    PRODUCTS_SOLD
---------------------------------------------------------------------------------------------------------------- -------------
Gaming                                                                                                                   7
Accessories                                                                                                              4
Electronics                                                                                                              3
Mobiles                                                                                                                  2
```

3. Highest revenue-generating category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 13      SELECT MAX(CATEGORY_REVENUE)
 14      FROM (
 15          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 16          FROM CATEGORY C2
 17          JOIN PRODUCT P2
 18              ON C2.CATEGORY_ID = P2.CATEGORY_ID
 19          JOIN ORDER_DETAILS OD2
 20              ON P2.PRODUCT_ID = OD2.PRODUCT_ID
 21          GROUP BY C2.CATEGORY_ID
 22      )
 23  );

CATEGORY_NAME                                                                                                    TOTAL_REVENUE
---------------------------------------------------------------------------------------------------------------- -------------
Gaming                                                                                                                   34800
```

4. Average sales per category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      ROUND(AVG(OD.QUANTITY * OD.PRICE), 2) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY AVERAGE_SALES DESC;

CATEGORY_NAME                                                                                                  AVERAGE_SALES
---------------------------------------------------------------------------------------------------------------- -------------
Gaming                                                                                                                  8700
Electronics                                                                                                             4750
Mobiles                                                                                                                 3600
Accessories                                                                                                          3133.33
```